Chakole Ayush Anandrao

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.18
adarsh sanskar vidhyalaya	Class XII (state board of maharashtra)	2023	79.83%
shri niketan madhyamik vidhyalaya	Class X (state board of maharashtra)	2021	87.80%

TECNICAL SKILLS

Languages: C, C++, JavaScript, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Web: React.js, Node.js, Express.js, HTML, CSS, JavaScript

Database: MySQL

Tools: Git, GitHub, VS Code

RELEVANT COURSES

Machine Learning for Process Systems Engineering, **Computer Vision**, Numerical Methods, Elementary Linear Algebra, Applied Mathematics, Differential Equations, Calculus

PROJECT

CodeVault - DSA Performance Analytics & Recommendation Platform | C++, React.js, Node.js, Express.js, MySQL

GitHub: https://github.com/ayushchakole100-blip/CodeVault

- Developed a **full-stack DSA performance analytics platform** to track problem submissions, acceptance rates, coding streaks, topic-wise performance, and personalized learning goals.
- Engineered a **C++ recommendation engine using STL and priority queues** to rank DSA problems based on topic weakness, difficulty alignment, and user rating, generating personalized practice recommendations.
- Built **RESTful APIs with Node.js and Express.js** and integrated **MySQL** for relational data management across users, problems, submissions, topics, and goals; implemented **JWT-based authentication and protected routes**.
- Designed a responsive **React.js analytics dashboard** with real-time performance metrics, activity visualizations, difficulty analysis, weak/strong topic identification, and goal progress tracking.

live-whiteboard | React.js, Node.js, Express.js, Socket.io, WebSockets, Firebase Authentication

GitHub: https://github.com/ayushchakole100-blip/live-whiteboard

Live Link - https://live-whiteboard-mu.vercel.app

- Engineered a **full-stack collaborative whiteboard** with **React, Node.js, and WebSockets**.
- Applied advanced **algorithmic state management** to synchronize high-frequency drawing events, achieving **sub-millisecond latency** across isolated **WebSocket channels**.
- Solved session desynchronization by hydrating late joiners with **in-memory state history**.
- Implemented **continuous coordinate snapshotting** to dynamically render shape primitives, significantly minimizing network payload before committing **final vectors to the server**.
- Secured user workspaces using **Firebase authentication** and **dynamic URL route-guarding**.

TECHNICAL PROFICIENCY AND PRACTICE

Data Structures & Algorithms and Competitive Programming | C++

Solved 500+ DSA problems across **LeetCode** and **Codeforces**, reinforce algorithms, data structures, problem-solving.

Implemented optimized **C++** solutions with focus on **time-space complexity** and competitive programming.

Applied graph algorithms, dynamic programming, greedy methods, trees, and STL in coding challenges.

Full-Stack Web Development | Frontend, Node.js, DBMS & SQL

- **Built** responsive web applications using **React.js, HTML, CSS**, and **JavaScript** with reusable UI components.
- **Developed** backend services using **Node.js** and **Express.js**, implementing secure **RESTful APIs**.
- **Designed MySQL** databases, optimized **SQL queries**, and managed version control using **Git** and **GitHub**.

POSITIONS OF RESPONSIBILITY

Departmental Representative | Diesta, IIT Hyderabad - *(2024-2025)*

Represented the Chemical Engineering department in Diesta, the inter-departmental competition of IIT Hyderabad.

Coordinated participation of department students in various events and facilitated smooth communication between the

department and organizers.

Sponsorship Coordinator | TEDx IITH - *(2024-2025)*

Worked as a Sub-cord in TEDx IITH, responsible for identifying and reaching out to potential sponsors for the event.

Collaborated with the core team to draft sponsorship proposals and managed communications with external organizations

to secure funding.

EXPERIENCE

Mentor | Physics Wallah (PW), Online - *Aug 2024 - Oct 2024*

Mentored JEE aspirants in **Physics, Chemistry, and Mathematics**, conducting doubt-solving sessions and guidance.

Improved student understanding through structured explanations and personalized problem-solving strategies.

EXTRACURRICULAR

- Member of the IIT Hyderabad Cricket NSO; actively participate in cricket, gym training and other athletic activities, demonstrating discipline, teamwork and commitment to fitness.

ACHIEVEMENT

- Solved 500+ DSA/competitive programming problems across LeetCode and Codeforces, demonstrating strong problemsolving and optimization skills.